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Static Voltage Stability Criterion of Renewable Energy Sources based on Power Flow Analysis

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***Abstract*—With the continuous increase in the penetration rate of new energy sources, the strength of the power grid gradually weakens, and the anti-interference capability of the system also decreases accordingly. Therefore, the real-time identification of the static voltage stability of power systems is of particular importance. The traditional approach for identifying the static voltage stability boundary of power systems is the *P-V* curve method. However, this method exhibits high computational complexity and strong dependence on the operational scenarios of the system, making it difficult to meet the requirements of real-time monitoring under complex scenarios. In view of this, this paper proposes a static voltage stability assessment method based on the singular points of the Jacobian matrix, and theoretically proves the effectiveness of this novel voltage stability assessment method. The reliability of the proposed method is verified through simulations.**

***Keywords*—Grid-following converter, Control strategy, Static voltage stability, Jacobian matrix.**

I. INTRODUCTION

With the continuous increase in the penetration rate of renewable energy source (RES), the number of voltage source converters (VSC) connected to the power grid has been increasing progressively, which leads to a gradual weakening of grid strength[1]-[3]. Concurrently, the system's capability to resist disturbances decreases accordingly, and the interaction between inverters and the AC power grid may give rise to unstable fluctuations in grid voltage [4]. To accurately evaluate system fluctuations, proactively identify the risks of static voltage instability, and optimize power grid structure as well as equipment configuration, the assessment of static voltage stability assumes crucial significance.

At present, two primary methods are widely applied to the calculation of static voltage stability in power systems: the *P-V* curve method [5]-[6] and the *Q-V* curve method [7]-[8]. The former can intuitively reflect the stability margin but requires a large number of power flow calculations. The latter, on the other hand, can clearly demonstrate the robustness of node voltage to reactive power but only one node can be considered at a tim. Therefore, both of the aforementioned methods have limitations in engineering applications.

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This paper proposes a novel method for determining the static voltage stability of power systems by analyzing the singular points of the Jacobian matrix of power flow equations. Unlike the traditional *P-V* curve method, which relies on complex continuous power flow iterative calculations, this method directly solves the critical voltage amplitude using the line impedance modulus and the new energy output power modulus. This significantly simplifies the calculation process of critical voltage and enhances the capability of rapid identification of the static voltage stability boundary.

II. VSC GRID-CONNECTED SYSTEM MODELLING

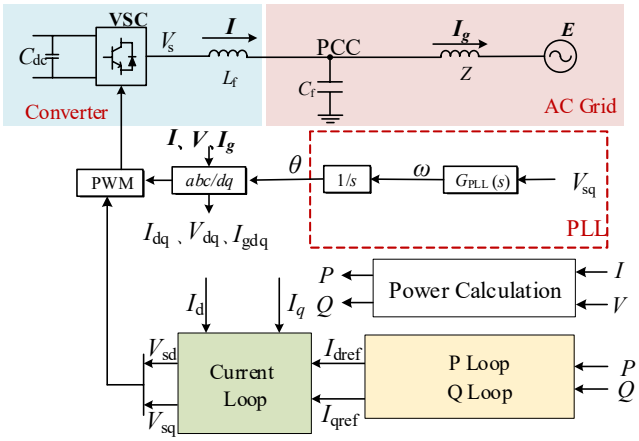


Fig. 1. VSC grid-connected system and its control diagram.

Most grid-connected renewable energy generation systems integrated with inverter technology rely on Phase-Locked Loops (PLLs) to achieve grid synchronization, which constitutes the most widely adopted control strategy in current engineering practice.. A typical VSC grid-connected system including its primary circuit and control structure, is shown in Fig. 1. In this system, $\boldsymbol{V_s}$ represents the voltage vector at the inverter output, L_f and C_f are the filtering inductance and capacitance, respectively, C_{dc} is the DC-side capacitor, Z is the equivalent impedance of the grid, $\boldsymbol{I_g}$ denotes the current vector flowing through Z, $\boldsymbol{E}$ is the voltage phasor of the infinite bus, and $\boldsymbol{V}$ and $\boldsymbol{I}$ represent the voltage and current phasors at the Point of Common Coupling (PCC), respectively. Furthermore,

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in the analysis presented in this paper, the subscripts '*d*' and '*q*' represent the *d*-axis and *q*-axis components of the corresponding quantities in the *dq* reference frame, while the subscript 'ref' denotes the reference values of the corresponding quantities.

A. Phase-Locked Loop

When VSC grid connected system is subjected to small disturbance voltage phase angle changes, PLL dynamically adjusts the *q*-axis voltage component at PCC point. V_q is used to maintain voltage stability. The typical phase-locked loop control principle is shown in Fig. 2.

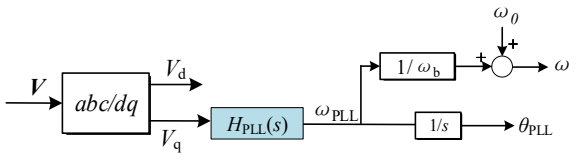


Fig. 2. Typical PLL control schematic.

In Fig. 2, ω_b is the reference value of 100π. The ω_{PLL} represents the rate of change in rotational speed between the *dq* coordinate system and the global *xy* coordinate system; θ_{PLL} is the output phase angle of PLL; ω is the PLL output phase frequency and $\omega=\omega_{PLL}/\omega_b+\omega_0$; s is the Laplacian operator; $H_{PLL}(s)=k_{PLLp}+k_{PLLi}/s$ is the transfer function of the phase-locked loop PI controller; k_{PLLp} and k_{PLLi} are proportional coefficients and integrals, respectively coefficient; select ω_{PLL} and θ_{PLL} as PLL state variables.

The PLL state equation in *dq* coordinate system is shown in (1).

$$\begin{cases} \dfrac{\mathrm{d}\omega_{\mathrm{PLL}}}{\mathrm{d}t} - k_{\mathrm{PLLp}}\dfrac{\mathrm{d}V_q}{\mathrm{d}t} = k_{\mathrm{PPLi}}V_{\mathrm{q}} \\ \dfrac{\mathrm{d}\theta_{\mathrm{PLL}}}{\mathrm{d}t} = \omega_{\mathrm{PLL}} \end{cases} \tag{1}$$

B. Inner Current Control Loop

The current inner loop control diagram of VSC grid connected system is shown in Fig. 3.

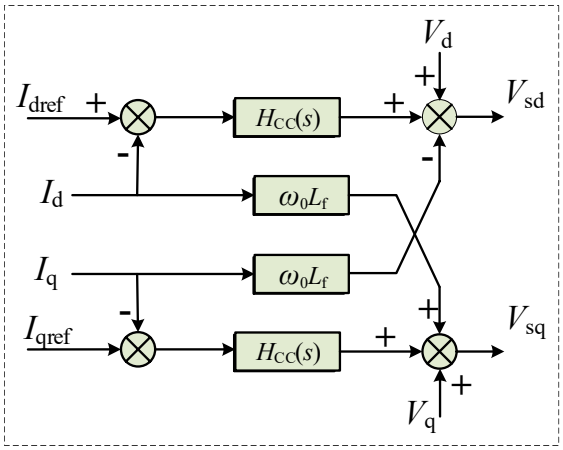


Fig. 3. Inner current control loop schematic.

In Fig. 3, $H_{CC}(s)=k_{CCp}+k_{CCi}/s$ is the transfer function of the current inner loop PI controller; k_{CCp} and k_{CCi} are the proportional coefficient and integral coefficient of the PI controller, respectively; Define x_1 and x_2 as the integrator outputs of the *dq* axis current inner loop PI controller, respectively.

The dynamic equation of PCC point voltage V is shown in (2).

$$\begin{cases} V_{\mathrm{sd}} = V_{\mathrm{d}} + x_1 + k_{\mathrm{CCp}}\left(I_{\mathrm{dref}} - I_{\mathrm{d}}\right) - \omega_0 L_{\mathrm{f}} I_{\mathrm{q}} \\ V_{\mathrm{sq}} = V_{\mathrm{q}} + x_2 + k_{\mathrm{CCp}}\left(I_{\mathrm{qref}} - I_{\mathrm{q}}\right) + \omega_0 L_{\mathrm{f}} I_{\mathrm{d}} \end{cases} \tag{2}$$

Selecting x_1 and x_2 as current loop state variables, the current loop state equation of the VSC grid connected system in the *dq* coordinate system is shown in (3).

$$\begin{cases} \dfrac{\mathrm{d}x_1}{\mathrm{d}t} = k_{\mathrm{CCi}}\left(I_{\mathrm{dref}} - I_{\mathrm{d}}\right) \\ \dfrac{\mathrm{d}x_2}{\mathrm{d}t} = k_{\mathrm{CCi}}\left(I_{\mathrm{qref}} - I_{\mathrm{q}}\right) \end{cases} \tag{3}$$

C. Power Loop

The power outer loop control diagram of VSC grid connected system is shown in Fig. 4.

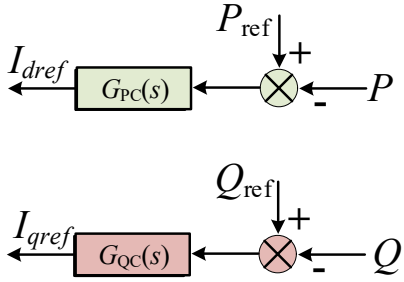


Fig. 4. Power outer loop control diagram.

The forms of reactive power control loop and active power control loop are similar, so their dynamic equations can be combined and written in the form shown in (4).

$$\begin{aligned} I_{\mathrm{dref}} &= G_{\mathrm{PC}}(s)\left(P - P_{\mathrm{ref}}\right) \\ I_{\mathrm{qref}} &= G_{\mathrm{QC}}(s)\left(Q - Q_{\mathrm{ref}}\right) \end{aligned} \tag{4}$$

Among them, $G_{PC}(s)=k_{PCp}+k_{PCi}/s$ is the transfer function of the active power outer loop PI controller. $G_{QC}(s)=k_{QCp}+k_{QCi}/s$ is the transfer function of the reactive power outer loop PI controller.

III. Voltage Stability Analysis of New Energy Grid Connected System

To reduce the dimensionality of complex power grid models and ensure that this method can adapt to new energy grid-connected scenarios with different topological structures, this study simplifies the external power grid into a single-machine infinite bus system, where the voltage amplitude is set to ***E*** and

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the phase angle is 0.The new energy grid-connected system is connected to the slack bus through an equivalent impedance $Z=R+jX$. In the model, the influence of the equivalent impedance of the circuit on the resistance R during power transmission is considered, which effectively improves the accuracy of the weak power grid model. The port voltage amplitude of the new energy grid-connected system is $\boldsymbol{U}$, and the phase angle is θ.

In order to meet the engineering reality of using new energy sources such as wind power and photovoltaics as transmission power sources, and to distinguish them from traditional load models (which absorb power), the new energy grid connected system is defined as the power source role (the inverter side node is the *P-Q* node), and the corresponding power equation is shown in (5).

$$\begin{aligned} -(P+jQ) &= U\frac{\hat{E}-\hat{U}}{R-jX} \\ &= \left(U\cos\theta + jU\sin\theta\right)\left(\frac{E-U\cos\theta+jU\sin\theta}{R-jX}\right) \end{aligned} \tag{5}$$

In the formula, P is the active power generated by the new energy grid connected system, Q is the reactive power generated by the system, R is the resistance value of the line, and X is the reactance value of the line.

By separating the real and imaginary parts of Formula (5), we can obtain:

$$\begin{aligned} P &= -\frac{UE\left(R\cos\theta - X\sin\theta\right)-U^2R}{R^2+X^2} \\ Q &= -\frac{UE\left(R\sin\theta + X\cos\theta\right)-U^2X}{R^2+X^2} \end{aligned} \tag{6}$$

According to (6), constructing Jacobian matrix by taking partial derivatives of active power P and reactive power Q in the power flow equation with respect to phase angle θ and voltage amplitude U:

$$\begin{bmatrix}\Delta P\\ \Delta Q\end{bmatrix} = \begin{bmatrix} \frac{UE\left(R\sin\theta+X\cos\theta\right)}{R^2+X^2} & \frac{UE\left(R\cos\theta-X\sin\theta\right)-2U^2R}{R^2+X^2} \\ \frac{UE\left(R\cos\theta-X\sin\theta\right)}{R^2+X^2} & \frac{UE\left(R\sin\theta+X\cos\theta\right)-2U^2X}{R^2+X^2} \end{bmatrix}\begin{bmatrix}\Delta\theta\\ \Delta U/U\end{bmatrix} \tag{7}$$

Further simplify the Jacobian matrix to obtain:

$$\begin{bmatrix}\Delta P\\ \Delta Q\end{bmatrix} = \begin{bmatrix} \frac{U^2X}{R^2+X^2}+Q & \frac{U^2R}{R^2+X^2}-P \\ -P-\frac{U^2R}{R^2+X^2} & \frac{U^2X}{R^2+X^2}-Q \end{bmatrix}\begin{bmatrix}\Delta\theta\\ \Delta U/U\end{bmatrix} \tag{8}$$

Let the determinant of (8) be 0, and obtain (9):

$$\frac{U^4}{R^2+X^2} - P^2 - Q^2 = 0 \tag{9}$$

Equation (9) indicates that at the singular point of the Jacobian matrix, the resistance R cannot act alone, but rather the impedance mode and power mode/capacity function. Furthermore, at the critical point, derive the formula for calculating the critical voltage amplitude based on the singular point condition of the Jacobian matrix as follows:

$$U_{\text{cri}}^2 = \sqrt{R^2+X^2}\sqrt{P^2+Q^2} \tag{10}$$

Among them, U_{cri} represents the critical voltage of the singular point of the Jacobian matrix.

By substituting the two equations in (6) and summing the squares, the equation for the power circle can be expressed as:

$$\left(-P+\frac{U^2R}{R^2+X^2}\right)^2 + \left(-Q+\frac{U^2X}{R^2+X^2}\right)^2 = \frac{U^2E^2}{R^2+X^2} \tag{11}$$

By substituting the critical voltage expression of formula (10) into formula (11), the expression satisfied when the critical voltage is stable is obtained as follows:

$$E = \sqrt{2}\sqrt{\sqrt{R^2+X^2}\sqrt{P^2+Q^2}-QX-PR} \tag{12}$$

According to this equation, we can make $\Delta = \sqrt{2}\sqrt{\sqrt{R^2+X^2}\sqrt{P^2+Q^2}-QX-PR}$. Δ can be used as a new voltage stability criterion for load and weak *PQ* as power sources. Calculate the voltage stability criterion Δ and perform static voltage stability discrimination. If $\Delta > E$, it indicates that the system's static voltage is stable; If $\Delta < E$, it indicates that the system's static voltage is unstable; When $\Delta = E$, it indicates that the system is in a static voltage critical stable state. The present invention uses this method to replace the consideration of *P-V* curve (nose curve) with the consideration of the power circle of Jacobian matrix singular points. Avoiding the problem of complex calculations for continuous power flow in *P-V* curves and simplifying the assessment of system static voltage stability.

Compared with the PV curve method, this proposed method can significantly reduce the computation time. For a simplified system with a single line, the single computation time of this method ranges from 10 nanoseconds to 5 microseconds, whereas the single computation time of the *PV* curve method (per P_k) is between 50 and 100 microseconds, representing a time reduction of approximately 98.7%.For complex systems with multiple nodes and multiple lines, the simplified criterion is extended to the "comprehensive criterion for multi-line systems" (e.g., the criterion value is calculated for each line, and the minimum value is adopted as the system stability index). Assuming there are 10 lines, the single computation time of the extended simplified criterion method is 100 nanoseconds to 50 microseconds, while that of the *PV* curve method is 500 to 1000 microseconds, corresponding to a time reduction of about 93.3%.In conclusion, whether for a simplified single-line system or a complex multi-node system, the power flow calculation method based on the simplified criterion proposed in this paper demonstrates disruptive advantages in computation time compared with the traditional *PV* curve method.

The flow chart of voltage stability discrimination is shown in Fig. 5.

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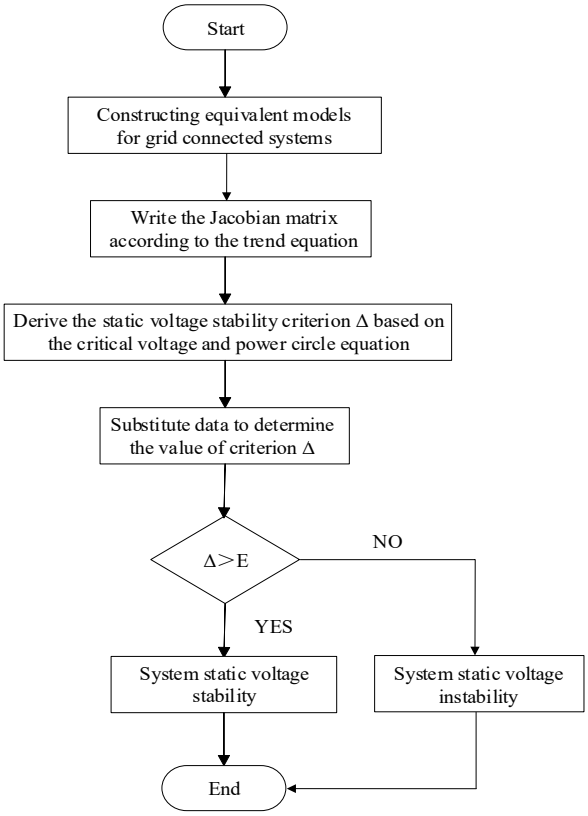


Fig. 5. Voltage stability discrimination flow chart.

IV. Simulation Verification

To demonstrate the effectiveness of the proposed method, a VSC mathematical model and its grid-connected system, as shown in Fig. 1, were built in Matlab/Simulink, with the specific simulation parameters are provided as follows:

TABLE 1 VSC GRID-CONNECTED SYSTEM PARAMETERS

Symbol	***Description***	***Value* (p.u.)**
E	Voltage of an infinite power grid	1
P_{ref}	Reference Values for Active Power	0(t<1), 0.5*(t-1) (t>1)
Q_{ref}	Reference Values for Reactive Power	0.4
L_f	Filter Inductor	0.05
R, X	Equivalent resistance and reactance of AC power grid	0.08, 0.5
$H_{PLL}(s)$	Phase-Locked Loop (PLL) PI Controller	10+50/s
$H_{CC}(s)$	Current Loop PI Controller	0.3+10/s
$G_{PC}(s)$, $G_{QC}(s)$	Active Power and Reactive Power Loop PI Controller	0.5+30/s

By changing the active power on the load side to Pref=0.5 * *t*, where *t* is time (seconds), we simulated the continuous increase in active power output of the grid connected system and observed the corresponding changes in terminal voltage. The results are shown in Table 2.

TABLE 2 CHANGE THE RESULT OF THE TERMINAL VOLTAGE U OBTAINED BY CHANGING THE ACTIVE POWER P

Label	***P***	Δ	***V***
1	0.886	0.806	1.069
2	1.336	1.043	1.064
3	1.987	1.321	1.048
4	2.587	1.544	1.024
5	3.336	1.804	0.977
6	3.934	2.018	0.918

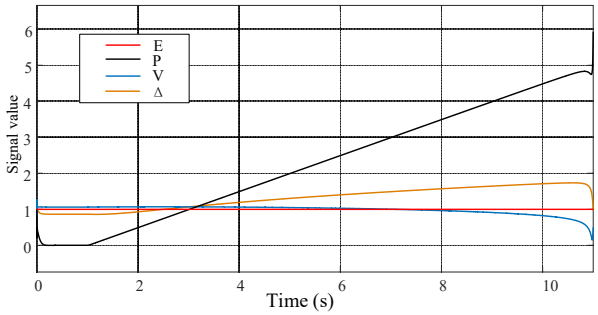


Fig. 6. Curve graph of P, V, E, Δ.

As can be seen, with other parameters unchanged, as the active output of the new energy grid connected system increases, the system terminal voltage *U* gradually decreases until it collapses. In the case of TABLE 2, criterion Δ is approximately close to the grid voltage *E*=1, which can be used as a critical stability condition. When criterion $\Delta > E$, the static voltage of the output system is stable. When criterion $\Delta < E$, the static voltage of the output system is unstable. Note again that this example is for load, weak *PQ* is for power supply, and attention should be paid to the flow direction of *P* and *Q*.

Mechanism of terminal voltage collapse: The continuous increase in active output of the output system leads to an increase in line current, and the reactive power loss of the line is proportional to the square of the current. However, the lack of reactive power support in the new energy grid connected system leads to a decrease in terminal voltage, which in turn requires a larger current to transmit the same active power, triggering a vicious cycle and ultimately forming voltage collapse, as shown in Fig. 6. When the voltage *U* is less than 0.9*E*, the criterion for voltage stability should be applied. When $\Delta > E$, the system is stable, and when $\Delta \le E$, the system becomes unstable. Combining the trend of the curve: 2 seconds ago, the active output of the new energy grid connected system was low, and the system had not yet entered a stable operating state. The voltage *U* was slightly higher than *E* (black), causing the value of Δ (blue) to approach *E*. 2 seconds later, the system was running normally, and the active output of the new energy grid connected system continued to increase. At this point, $\Delta > E$, indicating that the system meets the stable conditions and has not entered an unstable state. Although the voltage *U* at the grid connection point gradually decreases in the later stage, the increase in *P* exceeds the impact of *U* decrease (the dominant effect of *P* increase exceeds the weakening effect of

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U decrease), so the system remains stable. As P continues to increase, the line voltage loss ($PR+QX$) significantly increases and the system voltage margin decreases. In the later stage, P continued to increase, and after 10 seconds, U decreased to around $0.9E$, while the growth of Δ slowed down. After 11 seconds, Δ rapidly drops below E, and the voltage U also drops rapidly, causing voltage collapse. In summary, during the observation period, the new energy grid connected system established a normal operating state within 2 seconds, and the system met the static voltage stability criterion ($\Delta > E$) and was in a stable state from 2 to 11 seconds; But the voltage at the grid connection point continues to decrease with the increase of active power, and the voltage margin shrinks. After 11 seconds, the system did not meet the static voltage stability criterion ($\Delta \le E$) and experienced a voltage collapse.

When the P continues to increase as specified in Table 1, varying the Q allows for the investigation of the impact of reactive power on the system's voltage collapse time. Specifically, when Q is increased from 0.4 p.u. to 0.6 p.u. , the curves of P, V, E and Δ are illustrated in Fig. 7.

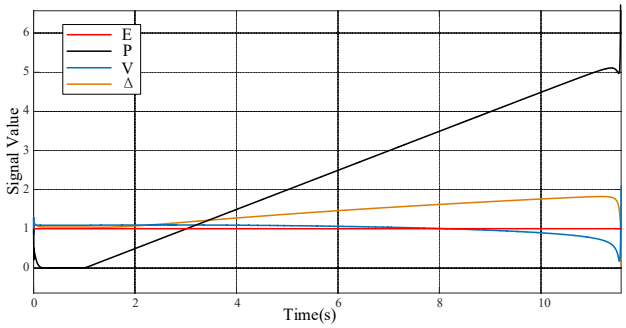


Fig. 7. Curve graph of P, V, E, Δ.at Q=0.6.

When the Q is increased to 0.6 p.u., it is observed that the system voltage collapse time is delayed by approximately 0.55 seconds, and the system's static voltage stability margin is significantly enhanced. The additional reactive power reserve can effectively offset the line voltage drop effect caused by the increase in P, thereby delaying the cascading process of static voltage instability. Ultimately, this manifests as a notable delay in the time interval from when the system approaches the stability limit to when voltage collapse occurs.

Considering the scenario where multiple renewable energy stations are connected to the grid, it is assumed that the parameters of VSC1 remain as presented in Table 1. VSC2 is connected in parallel with VSC1, and all parameters of VSC2 are identical to those of VSC1, except that its active power reference (P_{ref}) is increased to 1. This setup is intended to simulate a fault occurring in VSC1 under the condition of two parallel-connected VSC stations. The objective of this study is to verify whether the stability criterion based on Criterion Δ remains applicable, and the simulation results are presented in Fig. 7.

In comparison between Fig. 6 and Fig. 8, the values of both Criterion Δ and terminal voltage are nearly identical between the two scenarios. This indicates that under the condition of two parallel-connected VSC stations, when a fault occurs in one of the VSC stations, the proposed stability criterion remains accurate. Therefore, this criterion is applicable to multi-VSC parallel operation scenarios.

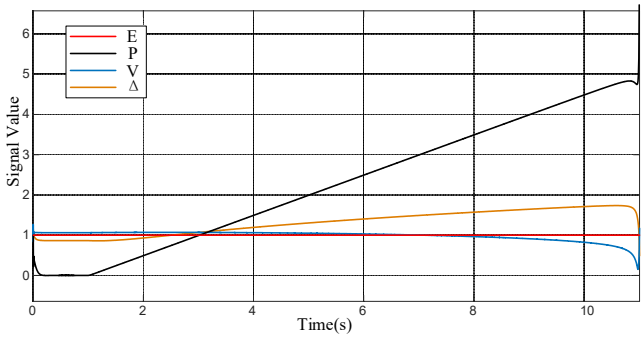


Fig. 8. P, V, E, Δ graph of the dual-VSC parallel system..

V. Conclusion

By analyzing the singular points of the Jacobian matrix determinant corresponding to the new energy grid-connected system, this study proposes a Voltage-Source-Converter -based voltage stability assessment method. This method eliminates the reliance of traditional P-V curve methods on continuous power flow calculations, which demand iterative solving of modified power flow equations. It simplifies the identification process of the static voltage stability boundary, improves computational efficiency, and enhances applicability in complex operating environments featuring fluctuations and intermittency of new energy output.

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